

Research article

Postponing Germany's nuclear phase-out: A smart move in the European energy crisis?

Dimitrios Glynos^{*}, Hendrik Scharf

Technische Universität Dresden, Faculty of Business and Economics, Chair of Energy Economics, D-01062, Dresden, Germany

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ABSTRACT

In response to the 2022–2023 energy crisis, the German government postponed the phase-out of the last three nuclear plants from the end of 2022 to the 15th of April 2023. Using the ELMOD and ELTRAMOD model cluster, we compare this decision with a counterfactual scenario without German nuclear capacity and derive its implications for the integrated European electricity market. The postponement of the nuclear phase-out reduced gas-fired power generation in Europe by 2.9 TWh, with a reduction of 1.6 TWh in Germany. The substitution of expensive power plants led to a decrease of almost €9 per MWh in average electricity prices in Germany. Furthermore, carbon dioxide emissions in Germany fell by 3.3 Mt. By extending our analysis to scenarios with increased nuclear capacity and different weather years, we illustrate the limitations of large capacity blocks for managing congestion in a decentralized energy system with multiple regional grid bottlenecks.

1. Introduction

The nuclear phase-out, a cornerstone of Germany's energy transition strategy, faced a pivotal moment during the energy crisis in 2022. Concerns over natural gas shortage prompted the German government to postpone the phase-out by three and a half months. As a result, the last three nuclear power plants in Germany were not phased out at the end of 2022 but instead continued to operate in a so-called "Streckbetrieb" (limited stretched-out operation) until the 15th of April 2023, using the remaining of their fuel elements.

In this study, we examine the multifaceted impacts of the nuclear phase-out postponement in Germany on various aspects of the energy landscape in 2023. Using a validated and calibrated model specifically for the period of the nuclear phase-out postponement, our analysis aims to contribute evidence-based insights to the ongoing debate in Germany by presenting factual data on the role of nuclear power during the energy crisis. Specifically, we assess the contributions to gas savings, electricity price reductions, and increased transmission adequacy in the critical period of 2023. The experience and lessons learned from this temporary strategy adjustment could serve as a reference for similar agile policies in future energy crises.

Additionally, we use this specific case study to draw general implications for the role of large generating units in systems with increasing penetration of decentralized low-carbon generators, such as intermittent renewables and batteries. The scenario results also provide insights into the potential reactivation of nuclear power plants in Germany, an option that has been raised repeatedly in the public debate.

In particular, following key research questions are being pursued:

1. *Security of supply and grid congestion:* To what extent have nuclear power plants contributed to securing energy supply by reducing gas-fired generation, given the risk of natural gas shortages? How has this decision reduced grid congestion, ensuring higher transmission security?
2. *Economic implications:* How did the postponement of the nuclear phase-out influence electricity prices in 2023? What trade-offs emerged for consumers, producers, and overall welfare in Germany and Europe? How does the available capacity of nuclear power affect the profitability of other technologies?
3. *Systemic perspective:* How has the postponement of the nuclear phase-out in Germany shaped the generation mix in the integrated European market? What conclusions can be drawn from the analysis of this extraordinary period of energy crisis for the central role of Germany in the European electricity market and the impact of its nuclear capacity in a system with a high share of renewable energy sources (RES)?

To answer these research questions, we explore four scenarios with different assumptions regarding the available German nuclear capacity from 1st of January to 15th of April 2023 utilizing the ELTRAMOD electricity market model and the ELMOD European power system model. We assess variations in generation mix, greenhouse gas (GHG) emissions and abatement costs, congestion management, power prices,

^{*} Corresponding author.

E-mail address: dimitrios.glynos@tu-dresden.de (D. Glynos).

Table 1

Deployability of nuclear power plants in the different scenarios analyzed. The REF scenario considers the actual availability in 2023, while the counterfactual scenarios that allow nuclear power generation assume the availability from 2021. The period is from January 1st to April 15th, respectively.

Data sources: [ENTSO-E \(2023\)](#) and [Federal Network Agency \(2023a\)](#).

Plant name	Installed net capacity ^a	Mean availability [%]			
		REF	Nuc ₀	Nuc ₃	Nuc ₆
Gemeinschaftskernkraftwerk Neckar	1310	53.7	0.0	98.3	98.3
Kernkraftwerk Emsland	1336	60.4	0.0	99.7	99.7
Kernkraftwerk Isar 2	1410	87.9	0.0	99.0	99.0
Kernkraftwerk Brokdorf	1410	0.0	0.0	0.0	99.9
Kernkraftwerk Grohnde	1360	0.0	0.0	0.0	98.0
Kernkraftwerk Gundremmingen	1288	0.0	0.0	0.0	98.7

^a Before shutting down.

welfare, and technology profitability across scenarios. Moreover, we extend our analysis to all weather years from 2012 to 2022 by considering the respective demand and RES generation profiles. The analysis derives eight key findings on the contribution of the nuclear capacity during the energy crisis and further implications for the role of large power generation units in systems with increasing penetration of decentralized low-carbon generators. The nuclear capacity scenarios are outlined below:

1. **REF:** Replicating the actual market clearing and grid situation using historical data and the reported available capacity for German nuclear plants, this factual scenario serves as the baseline for comparisons with the other scenarios.
2. **Nuc₀:** This counterfactual scenario assumes the phase-out proceeding as initially planned by the end of 2022. Hence, Germany's nuclear power capacity is set to zero.
3. **Nuc₃:** In this counterfactual scenario, we investigate the potential outcome of postponing the nuclear phase-out under the assumption that the last three power plants in Germany could operate without technical and fuel procurement constraints related to the impending phase-out. We refer to the availability conditions of nuclear power in 2021 as the last year without such restrictions.
4. **Nuc₆:** This counterfactual scenario explores a situation where Germany would have also postponed the previous phase of nuclear power plant shutdowns from 2021. In this scenario, six nuclear power plants in Germany would have been operational, with assumed availability factors based on the period from 1st of January to 15th April 2021, the last year in which all six plants were active.

As [Table 1](#) details, the three last active nuclear plants reported average availability between 53.7% and 87.9% between 1st of January and 15th of April 2023. In contrast, in 2021, all German nuclear power plants in operation had an average availability of more than 98.3% in this same period.

In the remainder of this paper, [Section 2](#) briefly outlines the models and the data used, along with the validation of both models, and [Section 3](#) presents the results, focusing on the security of supply and the economic impact of the decision to postpone the nuclear phase-out in Germany during the energy crisis. [Section 4](#) then contextualizes these findings and draws conclusions for the German and European power system.

2. Methodology to assess the role of nuclear power in the study period

2.1. Model description

The historical market behavior and the impact of the availability of nuclear capacity in Germany on the European integrated market in

the hypothetical scenarios are analyzed using the ELTRAMOD-ELMOD framework developed by the Chair of Energy Economics at TU Dresden. This dual-model architecture replicates the operational and economic optimization processes characteristic of European electricity markets and serves as a valuable tool for both historical analysis and strategic planning. The models have been extensively validated in numerous peer-reviewed papers and studies on RES integration, infrastructure and capacity investment, market design, and regulation or policy evaluation.

To determine the cost-optimal dispatch of electricity generation capacity and the necessary grid congestion management, the linear optimization models ELTRAMOD and ELMOD implement a two-stage market simulation similar to the mechanisms of the European electricity market.

In the first stage, the fundamental electricity market model ELTRAMOD determines the equilibrium between electricity supply and demand under perfect competition based on the marginal costs of the generation units. The objective function is the minimization of the total system costs, which in perfectly competitive markets aligns with maximizing individual generator profits and ensures optimal dispatch based on the merit order principle. ELTRAMOD features a detailed representation of power plants, incorporating economic and technological parameters such as efficiency ratios, emission factors, availability, cost structures, and fuel prices. It also considers operational constraints like must-run conditions of combined heat and power units, ramping thresholds for thermal plants, and reserve capacities, which are critical for realistic simulations of plant operations. The model integrates renewable energy generation time series and demand fluctuations, enhancing its accuracy and relevance to both current and future energy scenarios. Several storage options are implemented like reservoir and pumped-storage hydroelectric facilities, as well as battery storage systems, with a precise representation of the charging and discharging operations and storage levels. ELTRAMOD analyzes the fundamental interactions between generation capacities and demand with an hourly resolution over an annual cycle, capturing intertemporal dynamics in generation and storage under fluctuating feed-in and load conditions. Historical analysis is facilitated by modeling interconnections between European market areas using Net Transfer Capacities (NTC), which outline the maximum trading capacities across regions, assuming a “copper plate” approach of no internal bottlenecks similar to the current European market design.

In the second stage, ELMOD simulates the power flows resulting from the initial market clearing, adjusting power plant operations to mitigate grid congestion in a cost-minimizing way for the total system. ELMOD is a detailed, bottom-up model that features high-resolution mapping of the European transmission grid, incorporating regional demand shares and renewable capacities at each substation level. The model uses a direct current (DC) approximation for power flow analysis, which reduces the computational complexity while maintaining sufficient accuracy for operational planning. ELMOD accounts for the n-1 security criterion by incorporating a fixed Transmission Reliability Margin for each line. In case of overload, it determines the economic optimal adjustments in the generation profiles replicating the redispatch process implemented by the transmission system operations in Europe for managing grid congestion. With its comprehensive representation of transmission grid infrastructure, ELMOD can also be used to define flow-based capacity limits for electricity trade between market zones or as an infrastructure investment model within the integrated European electricity market. By simulating realistic market and grid operations, ELTRAMOD and ELMOD serve as critical tools for analyzing historical market behaviors and assessing future scenarios under various policy and market conditions.

For this study, we adapted the model to 2520 hours to represent the time horizon affected by the decision to extend the operation of the three German nuclear power plants. For the power flow analysis, we included all substations in Germany and all interconnection

points with its electricity neighbors as individual nodes while for model performance reasons, the national grids of all other regions and interconnection points between them are represented as one aggregated virtual node per region. For the virtual interconnection points between the other regions, the model considers the aggregated physical interconnection capacity.

A more detailed description of the existing database and the updated dataset used in the analysis as well as of the model accuracy in reproducing the historic behavior on the market and grid congestion is provided in the following sections. For further information on the dispatch-redispach models used and the mathematical formulation, the authors refer to Lieberwirth and Hobbie (2023) and Schönheit et al. (2020).

2.2. Input data

This section provides an overview of the dataset used for the analysis, explaining the sources and the pre-processing steps. For modeling the electricity market in hourly resolution for the period from 1st of January to 15th of April 2023, following data were essential: techno-economic power plant data, availability of the power plant fleet, load data, fuel prices, and the historical NTC values limiting the cross-border electricity exchanges within the European power grid. Model validation and calibration were based on historical Day-Ahead prices, generation profiles for different technologies, and historical congestion management demand represented by redispatch volumes.

The standard model database, documented by Kunze et al. (2019) and Lieberwirth and Hobbie (2023), provided the core data for the European power plant park and grid in our analysis. We meticulously cleaned and updated the power plant dataset to cover all operating power plants in the European market in the period from January to 15th of April 2023 and to align the aggregated installed capacity for different technology types with the data published on the Transparency Platform of the European Network of Transmission System Operators for Electricity (ENTSO-E, 2023). The most recent data for the power plant fleet in Germany was obtained from the list provided by the Federal Network Agency (2023a), the country's official electricity grid regulator. As Great Britain's generation capacity information is not available on the Transparency Platform, the data was downloaded from the Balancing Mechanism Reporting Service (BMRS) application programming interface (API) of Elexon Limited (2023).

Availability time series of RES were calculated as the ratio of actual generation in an hour to installed capacity. Feed-in data for PV, wind, and biomass capacities were acquired from ENTSO-E (2023) for most countries. For Great Britain, the aforementioned source was utilized. Italy's availability of renewable capacity was obtained from the official website of the Transmission System Operator (TERNA S.p.A., 2023).

Historic hourly load time series data spanning from 1st of January to 15th of April 2023, for European countries, were extracted from the Transparency platform of ENTSO-E (2023). Load data of Great Britain was provided again by the BMRS API through Elexon Limited (2023). The data of the island of Ireland was retrieved from the Transmission System Operator's official website EirGrid plc (2023). Similarly, actual generation time series needed to evaluate the model's quality were obtained from ENTSO-E (2023), except for Great Britain. Further time series required for model calibration and validation, such as Cross Border Commercial Schedule Exchanges for Electricity Exports and Day-Ahead Prices, were scrutinized from ENTSO-E (2023). The historic NTC data were also sourced from ENTSO-E (2023). When NTC were not published for Day-Ahead trade, Week-Ahead or Month-Ahead data were used, subsequently broken down to hourly resolution for the simulation spanning 1st of January to 15th of April 2023.

For all blocks in Germany exceeding an installed capacity of 100 MW, we used the actual historic availability based on the unavailability data derived from ENTSO-E (2023) in hourly resolution. For smaller power plants in Germany as well as for the power plants in the European

countries, we determined the hourly mean availability specific to each technology type within its respective country, based on the data from ENTSO-E (2023).

Fuel prices, critical for cost calculations, were obtained from various subscription platforms with commodity exchange data. Oil prices provided by CME Group Inc. (2023b) were already available on an hourly basis, while natural gas prices from European Energy Exchange (2023b) and coal prices based on CME Group Inc. (2023a) were available in monthly resolution and uniformly assigned to the hours within the respective months. Uranium and lignite prices were derived from Kunze et al. (2019) and remain constant throughout the analysis period. In addition to fuel prices, prices of European Emission Allowance from the European Energy Exchange (2023a) were imported as daily time series and assigned to the hours of each day of the analysis period.

To ensure that our grid model includes the latest developments in transmission expansion, we have updated ELMOD with the most recent static network model data from the European transmission system operators, provided by the European Joint Allocation Office (2023). The validation of the grid model is based on the quarterly report of the Federal Network Agency (2023b) as official authority and grid regulator in Germany. In the grid congestion report for the first quarter of 2023, which encompasses 85.7% of our analysis time horizon, the Federal Network Agency publishes data on positive and negative redispatch, curtailment of renewables and total grid congestion, as well as the Top 25 most overloaded grid elements. This allows a thorough validation, including the regional distribution of grid congestion in our model results.

2.3. Model validation

To assess the quality of a power system model, it is common to compare the model results of the base period with a couple of indicators derived from the statistics covering the same time horizon (cf. Hobbie et al., 2019; Lieberwirth and Hobbie, 2023; Schubert, 2016). The comparison of model results with the statistical data considers deviations in generation volumes, metrics of the power prices (averages, mean absolute error, correlation, standard deviation), and discrepancies of the model's ability to reproduce trade activities as well as positive and negative redispatch.

Fig. 1 gives an overview of the generation structure of Germany, some candidate countries and the whole area investigated. Overall, the model represents the generation volumes very well. The deviation of the German total power generation is 0.6% and the deviation of the modeled total generation of all countries considered from the actual sum is only 0.3%. As can be seen in Fig. 1, also the modeled shares of the single energy carriers in the candidate countries and the whole model region are well in line with the actual shares. This paper focuses primarily on the situation in Germany and therefore examines the other countries from a rather high-level perspective; the minor deviations are not considered critical since the model's ability to reproduce the German generation mix and the generation mix of the entire model area is highly satisfactory.

Also the modeled net exports of Germany match the real net scheduled exchanges of power pretty well. The deviation is 1.6 TWh only. As the literature shows deviations of this metric for full year simulations of up to 7.0 TWh (cf. Kunze et al., 2019, 2023), the deviation for the modeled period is considered to be in a reasonable range.

With an average value of €113.73 per MWh, the modeled German power prices match the average of the real Day-Ahead price time series for the German market zone. The standard deviation of the modeled German power prices is €43.50 per MWh. With €47.39 per MWh, the actual standard deviation is larger. The difference of €3.89 per MWh is a good outcome considering the range for the standard deviation discrepancy between modeled and real price time series reported in the literature, for instance Schubert (2016), who reproduces price time

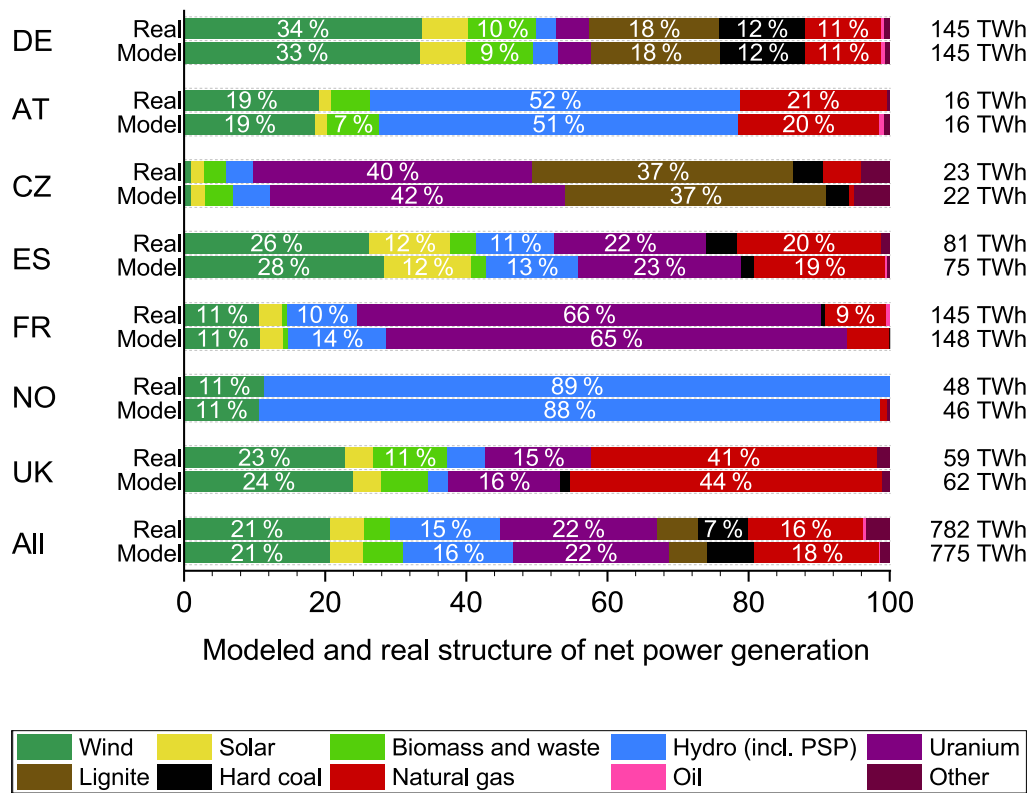


Fig. 1. Actual and modeled generation mix for candidate countries and the whole region considered.

series for 2012. Between the modeled unsorted prices and real Day-Ahead prices, the correlation coefficient is 82.8%, and 98.7% for the sorted prices. These values are higher than the correlation coefficients of Schubert (2016) and also of the values of other models (cf. Ladwig, 2018) and therefore a strong result as well. The mean absolute error (MAE) of the modeled sorted price duration curve from the actual one of the German power prices is with €4.10 per MWh also satisfying considering the mean absolute errors of other models that range between €2.28 and €5.93 per MWh (cf. Anke and Möst, 2021; Hobbie et al., 2019; Ladwig, 2018; Schubert, 2016).

To sum up, although the time horizon examined covers 28.5% of a normal model period challenging the model's ability to reproduce the actual conditions, the model shows a sufficient quality for the investigation intended within this paper. There are some discrepancies in relevant indicators when it comes to the results for some of the European countries modeled. Nevertheless, we anticipate that these discrepancies will have minimal impact on the scenario results due to the model's focus on the German electricity system and the total generation mix in Europe. The results for Germany, as well as for the entire modeled area, are highly consistent with the historic data.

The grid model produces reliable and realistic results that align well with figures from the official authorities. For the analysis period, the modeled positive and negative redispatch, and curtailment stand at 6.4 TWh, 3.3 TWh, and 4.3 TWh, respectively. The total grid congestion is estimated at 14 TWh, and its regional distribution is depicted in Fig. 2. In comparison, the Federal Network Agency's quarterly report, covering a period two weeks shorter than our analysis, reports positive redispatch at 5.2 TWh, negative redispatch at 2.6 TWh, RES curtailment at 3.6 TWh, with grid congestion reaching 11.4 TWh. Overall, despite the 16.7% longer reference period, the modeled grid congestion is only 22.8% higher, affirming the validity and accuracy of the model.

While the model may slightly overestimate grid congestion due to assumptions about thermal limits and the inability to replicate the real-world preventive congestion mitigation measures applied by the grid operators that reduce the total grid congestion, it effectively

reproduces the proportion and level of curative congestion measures. The visualization and comparison of model results with the Top 25 congested elements in Q1 2023 from the Federal Network Agency's report in Fig. 2 also underscores its ability to accurately identify main congestion corridors and the overloaded grid elements.

3. Eight key findings on the role of nuclear power on key parameters of the German and European power system

This Section presents the eight key findings of the contribution of nuclear power to savings of natural gas and other fossil fuels, grid adequacy, power prices, welfare, and GHG emissions in the critical early-year period of 2023. Moreover, it derives general implications for the role of large power generation units in systems with increasing penetration of decentralized low-carbon generators, such as intermittent renewable energy sources and batteries.

The findings are as follows:

1. German nuclear power in 2023 substituted 2.9 TWh of natural gas-fired power generation in Europe.

The postponed nuclear phase-out reduced natural gas-fired power generation by 1.6 TWh (−9.6%) in Germany and 1.4 TWh (−1.2%) in the other countries modeled. This outcome is derived from comparing the counterfactual scenario Nuc₀ with the REF scenario. Fig. 3 illustrates gas-fired power generation in Germany and Europe for all scenarios and weather years, providing insights into the potential for further gas savings and the effects of the available nuclear capacity in Germany.

The data point distribution of previous weather years suggests that the weather conditions in 2023 were a positive factor for natural gas savings, although the previous three years had been more advantageous in this regard. Under less favorable conditions, such as those in 2013 with lower weather-dependent RES generation, increased demand, and

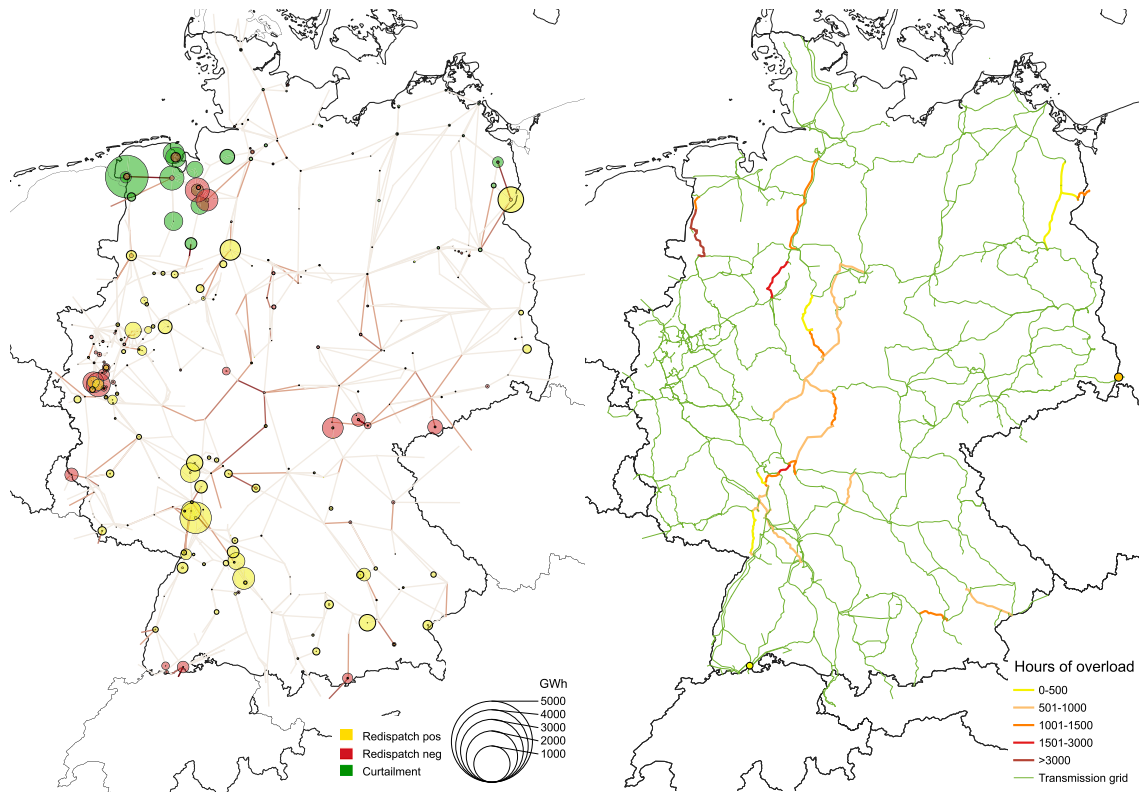


Fig. 2. Grid Congestion in Germany and Redispatch Activity in ELMOD on the left and visualization of the Top 25 congested grid elements in the first quarter of 2023 as published in the grid congestion report of the [Federal Network Agency \(2023b\)](#).

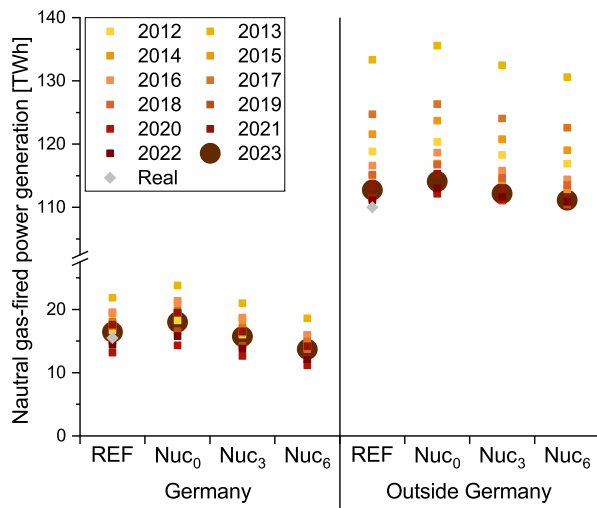


Fig. 3. Deviations in gas-fired generation of the counterfactual scenarios from the REF scenario for different weather years.

higher residual load, the impact of postponing the nuclear phase-out in Germany on natural gas savings would have been substantially higher. For this weather year, the total natural gas-fired generation in the REF scenario would have risen, ceteris paribus, by approximately 5.4 TWh in Germany and 20.6 TWh abroad. Also the difference in natural gas-fired generation in Europe between the Nuc₀ and the REF scenario results reaches its maximum at 4.2 TWh. At the same time, the particularly high additional gas-fired generation would have increased

the marginal risk of natural gas supply shortages. Therefore, every terawatt-hour saved by postponing the nuclear phase-out would have been even more crucial for the EU's efforts to ensure a stable natural gas supply.

2. Ensuring a more balanced distribution of capacity between north and south, the delayed phase-out reduced grid congestion by 1.6 TWh.

The postponed nuclear phase-out in Germany also had a positive impact on limiting grid congestion. The comparison of REF and the counterfactual scenario Nuc₀ in Fig. 4 reveals that without the nuclear plants, total congestion would have increased by 1.6 TWh (11.5%). Moreover, when compared to the REF scenario, the Nuc₀ scenario results show a total rise in overload hours of 15.2%, resulting in additional positive redispatch of 1.2 TWh (+18.9%) to correct the original market dispatch and ensure transmission viability.

The contribution of nuclear power plants in reducing grid congestion can be attributed to more evenly distributed capacities in market dispatch through higher capacity in western and southern Germany. When operating at full capacity as assumed in the Nuc₃, the last three nuclear power plants could have further relieved the main congestion points. In this scenario, grid congestion is reduced by an additional 1.3 TWh (−9.2%). In contrast, with all six nuclear power plants operational as in the Nuc₆ scenario, the grid congestion volume would be comparable to or slightly higher than in the REF scenario, depending on the weather year.

Contrary to the capacity adequacy on the European electricity market, weather years with more favorable conditions for renewable energy generation and lower residual load would have posed a greater challenge for transmission adequacy. Regarding security of supply, in terms of transmission adequacy, the realized configuration of nuclear power

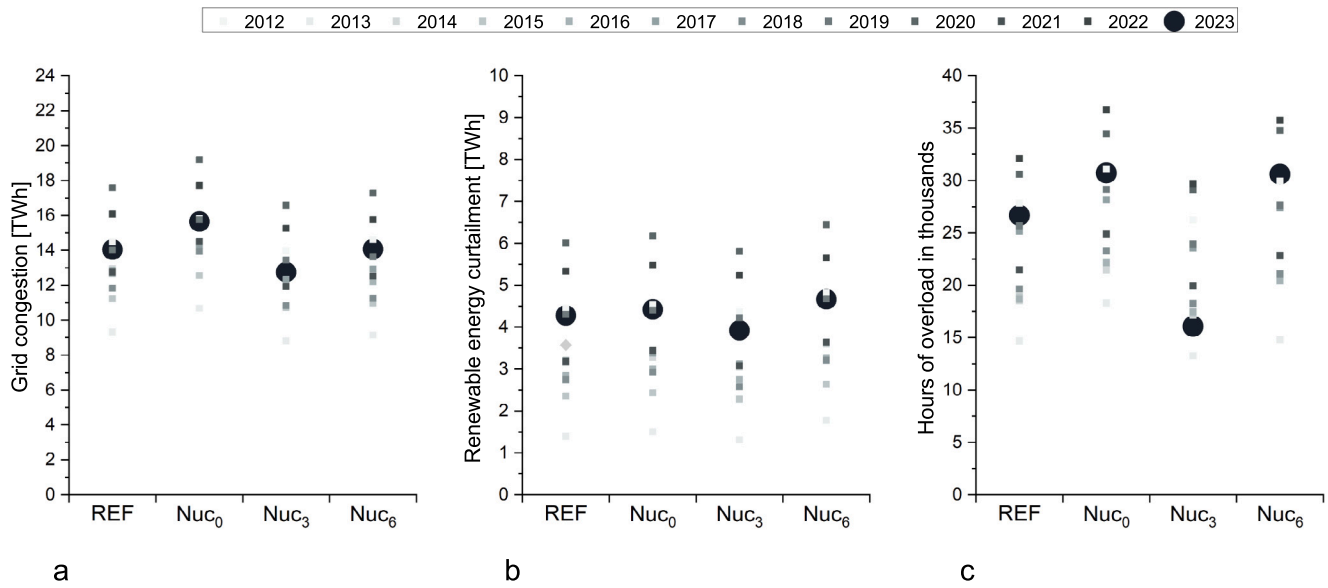


Fig. 4. Grid congestion (a), curtailment of RES as part of congestion management (b) and hours of overload (c) in the REF scenario and the counterfactual scenarios analyzed for the different weather years.

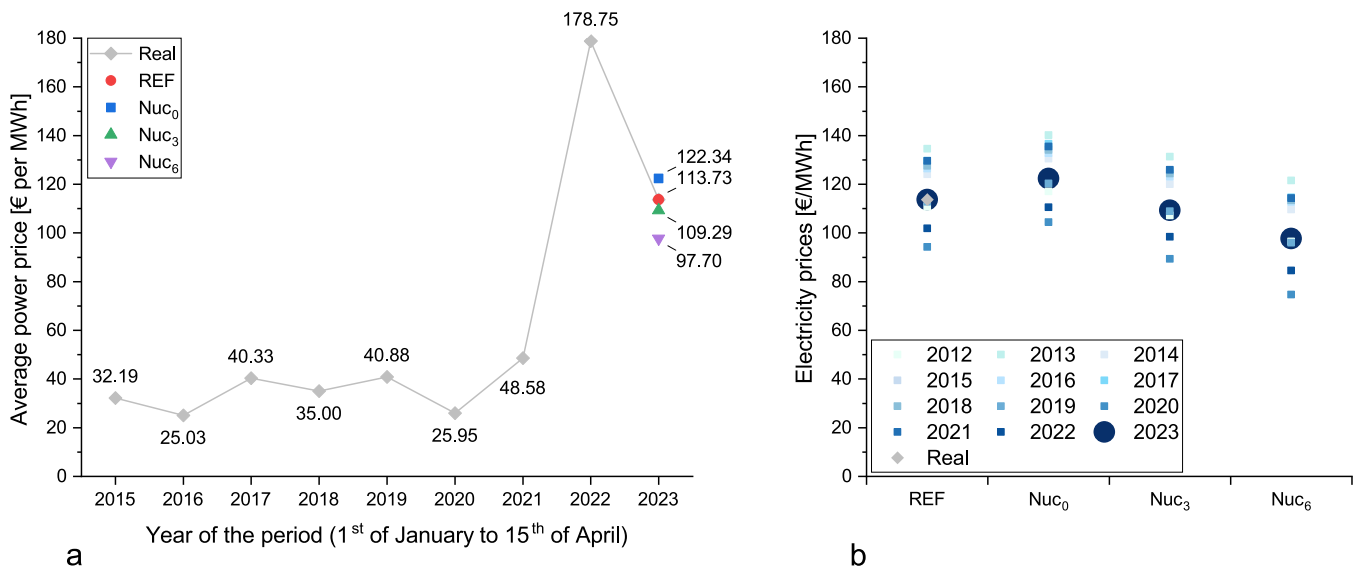


Fig. 5. Average power prices of the REF scenario and the counterfactual scenarios compared to historical Day-Ahead power prices (a) and the price spread for the other weather years (b).

plant capacity in Germany had a positive effect on reducing grid congestion compared to the counterfactual case of a 2022 phase-out, as analyzed in the Nuc₀ scenario. This positive contribution would have been realized for any weather year over the past decade, as highlighted by Fig. 4.

3. Average German power prices would have been almost €9 per MWh higher without the postponement of the nuclear phase-out.

The model results reveal that the decision to postpone the nuclear phase-out had a noticeable impact on power prices. Fig. 5 plots the average power prices of the different scenarios and contextualizes them with the average prices for the same period in the past, i.e., from 1st of January to 15th of April. Moreover, Fig. 5 depicts the results for the model runs using the demand time series and renewable availability profiles of the other weather years.

The historical average power price in Germany in the analysis period, as well as the price in the calibrated REF scenario of the model, was €113.73 per MWh. In the counterfactual Nuc₀ scenario, the average power price in Germany is at €122.34 per MWh and thus €8.61 per MWh higher than in the REF scenario, i.e., the postponement of the phase-out led to a price drop of 7.0%. The maximum potential price reduction from postponing the nuclear phase-out could have been another €4.44 per MWh greater, reaching a difference of €13.05 per MWh between the Nuc₀ and Nuc₃ scenario results, if the last nuclear power plants had operated at their reported available capacity for 2021.

As Fig. 5 shows, larger deviations in average power prices could have been realized in weather years characterized by high renewable energy generation and low residual load, such as 2020 or 2022. The residual load in 2023 was relatively low, meaning that the impact of the decision to extend the operation of the last three nuclear power plants on the reduction of the average electricity price was comparatively large.

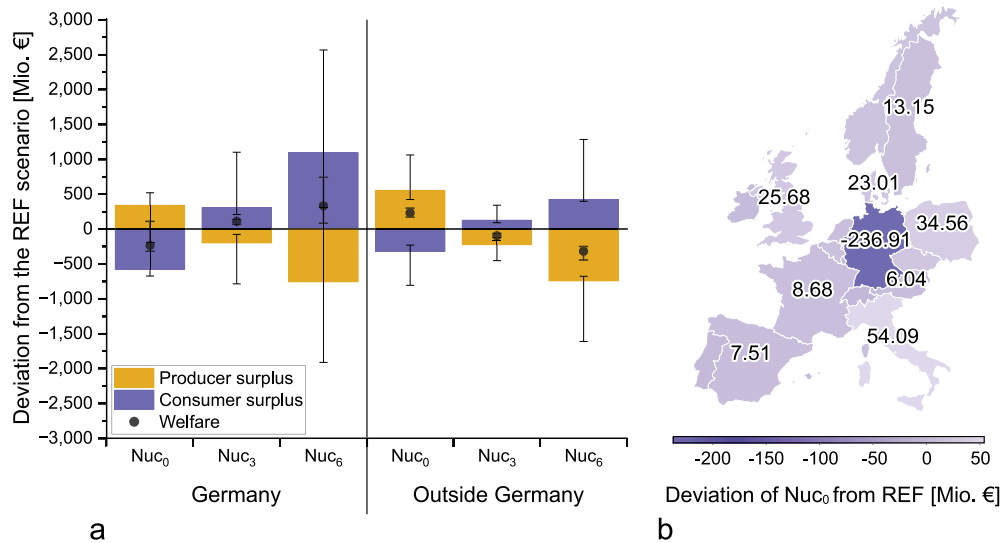


Fig. 6. Deviation of consumer surplus, producer surplus, and welfare, of the Nuc₀, the Nuc₃, and the Nuc₆ scenarios from the REF scenario in 2023 with the whiskers indicating the variations deduced from the model runs of the other weather years in Germany and the other regions modeled (a) and absolute deviation in welfare in Nuc₀ by country (b).

In contrast, in less favorable weather years the price level would generally have been higher, as more generation from expensive thermal power plants would have been required. The relatively small capacity of the nuclear power plants would not have been sufficient to displace expensive technologies from the merit order in many hours, narrowing the potential for price reduction through the postponement of the nuclear phase-out.

4. The postponement of the phase-out increased welfare in Germany by €236 Mio. in a redistribution of welfare at the expense of the European neighbors.

Fig. 6 shows the variation in producer surplus, consumer surplus, and welfare of the counterfactual scenarios and the reference scenario for Germany and the rest of Europe. In Germany, the delayed phase-out raised electricity consumers' surplus by €576 Mio. (+0.2%), while decreasing producers' surplus by €339 Mio. (-2.4%), resulting in a total welfare gain of €237 Mio. (+0.1%).

The impact on European welfare tends to be marginal. The decision of the German government to postpone the phase-out increased consumer surplus in all modeled European countries, notably in Luxembourg, which shares a market area with Germany, and in the neighboring Czech Republic and Austria. However, the overall welfare effect remained limited as gains on the consumer side were offset by losses on the producer side.

This dynamic is especially pronounced in countries like Czech Republic and Denmark. In the hypothetical Nuc₀ scenario, these nations expected to compensate for Germany's missing nuclear capacity by boosting revenues through exports. Nevertheless, as Germany reduced its import needs by postponing the nuclear phase-out, these additional revenues were not realized, leading to slight overall welfare losses for these countries.

Given the limited capacity of the three nuclear power plants, the postponement of the nuclear phase-out in Germany did not have a tangible impact on the merit order of its European neighbors by exporting cheap electricity from which more countries could profit. Instead, its contribution was to reduce expensive exports to Germany, with the common market zone of Luxembourg and Germany benefiting the most. In sum, the delayed phase-out reduced welfare by €231 Mio. in the rest of Europe, with the system-wide welfare gain being €6 Mio. only when also considering Germany.

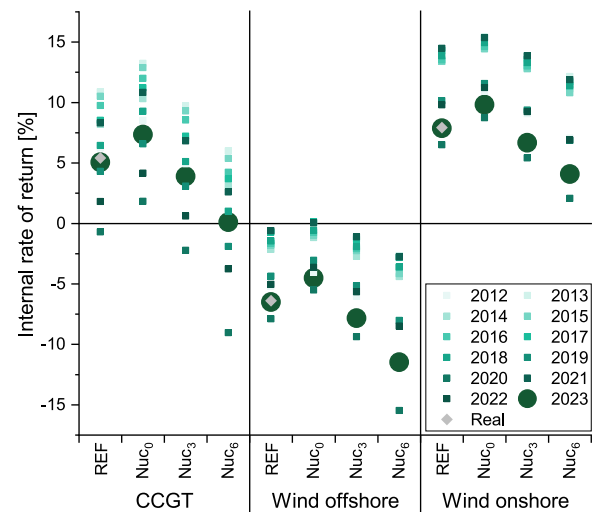


Fig. 7. Internal rate of return of candidate power generation technologies by scenario and weather year. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In analyzing the various price paths in our scenarios, we evaluate the possibility of windfall profits and whether the energy-only market could genuinely incentivize capacity investment at these increased energy crisis price levels. Investment decisions are made based on capital budgeting. Therefore, using the methodology described in [Anke and Möst \(2021\)](#), we quantify the effects of the different scenarios on the internal rates of return (IRR) for candidate technologies. The techno-economic data of the technologies are derived from reference indications for existing plants from the [Joint Research Centre \(2014\)](#). Fig. 7 shows the rates determined for each scenario and the distribution reflecting the impact of the weather year.

The nuclear phase-out policy significantly impacts the profitability of the exemplary plants in Germany. Assuming a capital cost rate of 10.0%, all reference plants are in the red in all scenarios. Consequently, the market conditions in the period analyzed do not seem to incite investments into new plants with market-based revenues only, even with the initially planned nuclear phase-out. The low return rates also rule out windfall profits for the technologies shown in Fig. 7 during

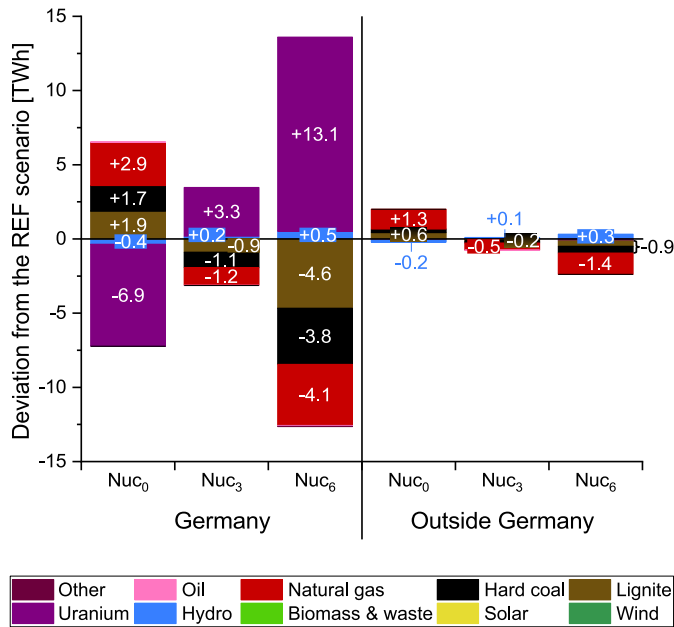


Fig. 8. Generation volumes for all technologies based on actual 2023 demand and weather profile.

the analyzed period. Onshore wind, however, could be an exception, especially under different weather conditions. The reference combined cycle and gas turbine (CCGT) plant could have had IRR clearly above 10.0% as well, but only in the Nuc₀ scenario and for the specific weather years 2013, 2015 or 2016.

5. Adding German nuclear capacity primarily reduces domestic coal-based generation and natural gas-based generation abroad.

Fig. 8 also breaks down the generation shifts in Germany and the rest of the model area and displays the variation in power generation by technology compared to the REF scenario. In the Nuc₀ scenario, Germany's domestic power plant fleet could have covered roughly two-thirds of the resulting 6.9 TWh supply gap arising from the nuclear phase-out in 2022. The remaining 1.3 TWh would have been supplied by additional imports from the European market.

The analysis of all scenarios illustrates how nuclear capacity in Germany primarily reduces domestic coal-fired generation and natural gas-fired generation in demand spike hours, but does not have the potential for significant shifts in the European electricity mix. In no scenario is the nuclear capacity in Germany sufficient to displace lignite or hard coal-based generation in Europe on a similar scale; instead, it primarily reduces import demand and the more expensive natural gas-fired generation in the European neighbors. It is important to note that a full replacement of gas or coal is not possible due to combined heat-and-power coefficients and other must-run conditions of thermal power plants used to cover remote heat demand.

6. Through the substitution of fossil fuel-fired plants, the additional nuclear capacity reduced GHG emissions in Germany by 3.3 Mt.

As shown in Fig. 9, the delayed nuclear phase-out reduced GHG emissions in Germany by 6.9% in the model period. Compared to the preliminary estimate of Huneke et al. (2023) for the emissions in 2023 of 177 Mt, the reduction amounts to only 1.9% of the annual emissions. In the rest of Europe, the results show a reduction in GHG emissions of 1.0 Mt, which represents a reduction of 1.0% of emissions over the analysis period.

The abatement costs shown in Fig. 9 have been calculated for the whole model area. Without the nuclear power plants in the Nuc₀ scenario, the abatement cost of reducing GHG emissions by one ton compared to REF would have been €88, while the scenarios with higher nuclear capacity would have further reduced both emissions and system costs. The marginal benefit, expressed by negative abatement costs, declines with increasing nuclear capacity, as less costly GHG-emitting power plants are gradually substituted by the additional nuclear capacity.

7. Additional nuclear capacity drives storage utilization.

Apart from fossil fuel-based generation, generation from hydro power plants varies across the scenarios. The deviations can be traced back to the activity of pumped-storage power (PSP) plants. In instances of electricity oversupply caused by the inflexibility of nuclear power plants, PSP stored excess electricity to later feed it into the grid during periods of high residual load. As a result, power generation from PSP plants consistently rises with the increasing availability of nuclear power.

The increased utilization of PSP in central and southern Germany, as well as neighboring countries in the Alps region and Luxembourg, together with the nuclear power plant capacity, ensured a more balanced distribution of generation between North and South, effectively reducing congestion in the North-South Transmission Corridor. Therefore, 0.1 TWh (+3.2%) less curtailment of RES to alleviate grid congestion was necessary in 2023 compared to the Nuc₀ scenario, with an additional reduction of almost 0.4 TWh (11.2% in total) being possible in the scenario Nuc₃.

8. The operation of all last six nuclear plants in Germany would have increased hours of overloads by 15%, exposing challenges for congestion management.

With six nuclear plants active, the increased capacity and the geographical distribution of the additional three nuclear power plants in Nuc₆ scenario would negatively impact grid congestion and increase curtailment of RES. Due to increased feed-in of nuclear power plants in Northwest, the Nuc₆ scenario has the highest curtailment, the highest number of overloaded lines, and more grid congestion compared to the REF scenario. As nuclear plants are dispatched at available capacity on the market (cf. Jenkins et al., 2018), they reduce generation from other controllable conventional plants in the North in the market dispatch, limiting the flexibility for negative redispatch prior to the congestion points. Furthermore, the geographic distance of nuclear power plants from the congestion points means that they cannot alleviate congestion by reducing their output, leaving curtailment of regional RES as the only congestion management option. Compared to the REF scenario, the Nuc₆ scenario shows an increase in curtailment of 9.1% of the actual RES curtailment corresponding to 0.4 TWh. This situation, exemplified by the Nuc₆ scenario, underscores the challenges of integrating RES in a system with large capacity blocks like nuclear power plants rather than more distributed flexible capacity.

To sum up, the analysis indicates positive effects of the nuclear phase-out postponement regarding power prices, grid congestion, consumer surplus, and fossil fuel savings. Negative effects were quantified for the producer side. While the impact of allowing the stretched-out operation of the last three nuclear power plants was comparatively small, considerably larger effects could have been achieved by increasing the availability of the existing or by reactivating nuclear reactors that had already been shut down, although the results indicate an increased volume of congestion management measures in the case of reactivation.

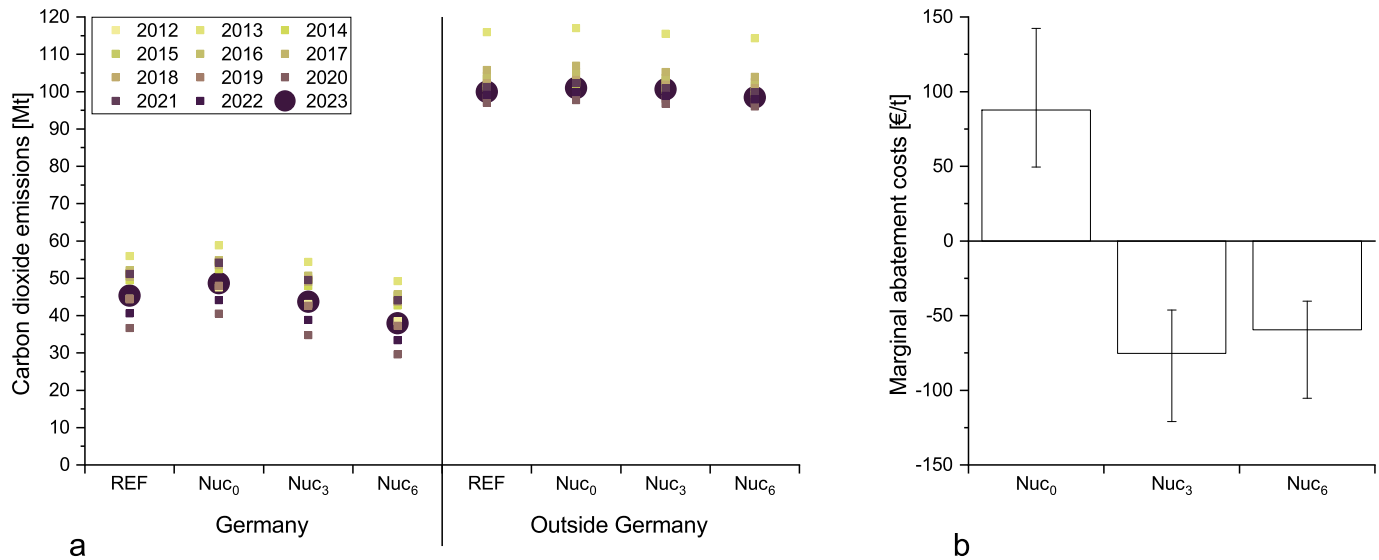


Fig. 9. GHG emissions by weather year (a) and marginal abatement costs of GHG emissions (b) in the model period.

4. Conclusion and implications

This paper analyzes the concrete policy decision of postponing the nuclear phase-out in Germany as a response to the energy crisis and its implications for power generation, natural gas savings, power prices, welfare, grid congestion, as well as GHG emissions and abatement costs. Its aim is to provide evidence-based facts and figures on this strategic adjustment, which can serve as a valuable reference point for future policies during an energy crisis. Extending the analysis with additional counterfactual scenarios of further nuclear capacity in Germany and alternative weather years, it allows long-term implications for the transformation paths of the energy system. These insights complement the direct lessons learned from the analysis of the decision to postpone nuclear phase-out during the 2022/23 energy crisis.

In retrospect, the decision to extend the operation of Germany's last three nuclear power plants was pragmatic. It did not only reduce natural gas-fired power generation in Germany by 2.9 TWh and by 4.2 TWh in Europe, but it also reduced the mean power price by €8.61 per MWh in Germany. The results vary across different weather years, with the realized weather year having a significantly greater impact than the variations in nuclear capacity across the scenarios. All results regarding the price level can be interpreted as a conservative estimation of the positive contribution of the phase-out postponement, as the scenarios do not account for the direct influence of gas savings on the gas spot market. The underlying assumption that marginal costs do not vary between the scenarios is well founded, as the decision to postpone the phase-out was taken only a month and a half before the period considered, so that we expect the energy companies to have already bought their commodities for this period as part of their risk management strategies. If power plant operators were more vulnerable to fuel price effects on the short-term gas market than considered, the increased gas consumption in the nuclear phase-out scenario could have resulted in even higher electricity prices than measured given the gas shortage in this period and assuming inelastic electricity demand.

As a result of the reduction in electricity prices, the analysis shows a marginal increase in the overall welfare in the European power sector. However, some countries had a negative effect on their welfare, as the loss in the producer surplus exceeded the gain in the consumer surplus. Accounting for the impact of gas savings in the electricity sector on the gas market in the quantification of the welfare effects could reveal a positive total welfare effect also in those countries.

A further positive effect of the postponement of the nuclear phase-out was the increase in security of supply. Besides natural gas savings contributing to higher capacity adequacy, the postponement also decreased grid congestion. These positive outcomes would have been realized for all other alternative weather years analyzed. In general, the realized weather year is found to be the most important factor for security of supply, with RES generation having a twofold effect. While higher RES generation, especially from wind, reduces natural gas consumption and improves capacity adequacy, it also increases grid congestion, compromising transmission adequacy.

Finally, the postponement of the nuclear phase-out contributed to meeting Germany's intersectoral annual GHG emission reduction target. It also led to a reduction in GHG emissions across Europe, although the reduction outside Germany was modest. Again, the realized weather year is a more significant driver of the GHG emission reduction potential than the analyzed variations in nuclear capacity in Germany. The results on GHG emission reduction should be interpreted as short-term abatement in the electricity sector, as the analysis does not account for intersectoral or intertemporal shifts. In other words, given the fixed volume of GHG emission allowances in Europe, the lifetime extension did not necessarily reduce long-term emissions in the European ETS sectors. To ensure a lasting impact, it may be necessary to remove the free allowances associated with the extended nuclear capacity from the market.

While postponing the nuclear phase-out in Germany proved to be a positive contribution to tackling the immediate energy challenges, this study also exposed potential limitations in aligning nuclear power with the characteristics of the future energy system. These are evident in increased storage utilization with increasing nuclear capacity and, more importantly, the challenges associated with managing regional grid congestion with an increasing share of intermittent renewable energy, as seen in the Nuc₆ scenario, where large centralized capacities provide limited options for mitigating the high number of regional grid congestion bottlenecks. This aspect should be considered in future decision-making regarding centralized blocks such as nuclear or coal capacity. Despite the high level of power prices observed in the analyzed period, our results align with ongoing discussions on the electricity market design, as the current energy-only market is found not to provide sufficient incentives for investments in new capacity.

Moreover, our results show that the integration of the European market is not so progressed, with national decisions on the commissioning or retirement of individual large capacity blocks having only a

marginal effect outside the home market. Given the progressing market integration, these cross-boarder dynamics will increase. Therefore, it can be concluded that similar national energy policy decisions in the future may have even larger potential negative impacts on neighboring countries than those identified in this study.

CRediT authorship contribution statement

Dimitrios Glynos: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hendrik Scharf:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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